# Developing For Cross Reality Interaction Loops

**B.Tech Thesis Project**

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## Introduction

Three-dimensional content creation remains one of the most skill-intensive tasks in modern computing. Traditional pipelines require hours of expert work in software like Blender or Maya, creating a barrier between people who *want* virtual objects and people who can *make* them. At the same time, consumer Mixed Reality (MR) headsets such as the Meta Quest 3 now place high-fidelity passthrough cameras, on-device neural networks, and hand-tracking in the same device — yet the interaction model for moving content *between* the physical and virtual remains underexplored.

This project asks a simple design question:

**Can a person bring a real-world object into a virtual workspace, manipulate it with their bare hands, edit it on a desktop, and return it to the headset — all without ever opening a file dialog?**

We answer in the affirmative through a working prototype that treats the *gaze* as the selection mechanism, the *hand* as the only input device, and the *real room* as the stage on which virtual objects behave. The contribution is not a new algorithm but a new **interaction loop**: a continuous, bidirectional pipeline where the boundary between physical, virtual, and desktop becomes invisible to the user.

## Literary Review

- **Reality–Virtuality Continuum.** Milgram and Kishino's foundational taxonomy [1] places Mixed Reality between fully real and fully virtual environments. Three decades later, consumer headsets finally have the sensors to make every point on this continuum reachable in a single device, but most XR applications still occupy a single point on it.
- **Natural User Interfaces.** Buxton's work on "natural" interaction [2] argues that the best interfaces *recede* — they leverage skills the user already has. Bowman et al. [3] formalise this for 3D interaction: pointing, grabbing, and throwing are gestures every human owns. Our design treats these as primitives rather than extending a 2D WIMP metaphor into 3D.
- **AI-Driven 3D Synthesis.** Recent advances in image-conditioned 3D generation — notably Meta's Segment-Anything-3D (SAM-3D) [4] and related diffusion-based methods [5] — have collapsed model-generation time from hours to seconds. These models are typically used in cloud workflows; their integration with live MR sensing remains rare.
- **Cross-Device XR.** Grandi et al. [6] and Marquardt et al. [7] explore handoffs between immersive and 2D devices, but most existing work treats the desktop as a *configurator* for the headset, not a peer in a round-trip workflow.

## Methodology

The system is organised around four design commitments — each a concrete answer to "how should this *feel*?"

3.1 Gaze-as-Selection

The user does not pick from a menu. The headset's on-device object detector continuously labels everything in view; the highest-confidence non-device label becomes the implicit prompt. Pressing a single controller button captures the corresponding region of the passthrough feed.

3.2 Cloud Synthesis as a Black Box

The captured image is dispatched to a hosted SAM-3D endpoint. To the user, this is a 20–30 s "loading" moment — to the system, it is the only step that requires more than millisecond-scale compute. The returned GLB is downloaded, parsed at runtime, and instantiated.

3.3 Real-Surface Physics

The instantiated mesh is wrapped with a collider, a rigidbody, and physics-based grab interactables (pinch, palm, controller, *and distance grab*). The MR Utility Kit's room-mesh provides surfaces for the object to land on — virtual physics, real geometry. The user can throw, place, and stack the digitised object alongside its physical counterpart.

3.4 Bidirectional Loop via a Shared Cloud

A single button transfers the active model's URL to a Realtime Database. A laptop service polls for new entries, downloads the GLB, and auto-opens it in Blender. After the user edits the asset in Blender, a one-click "Send to VR" panel uploads the revised file to Firebase Storage and writes a record to a reverse channel. The headset polls this channel and respawns the updated mesh — without restarting the app, without manual import, and without any explicit file naming.

Fig 1.1: Actual image of the cross reality application.

Fig 1.2: Opening the reconstructed model in blender seamlessly without needing to exit VR

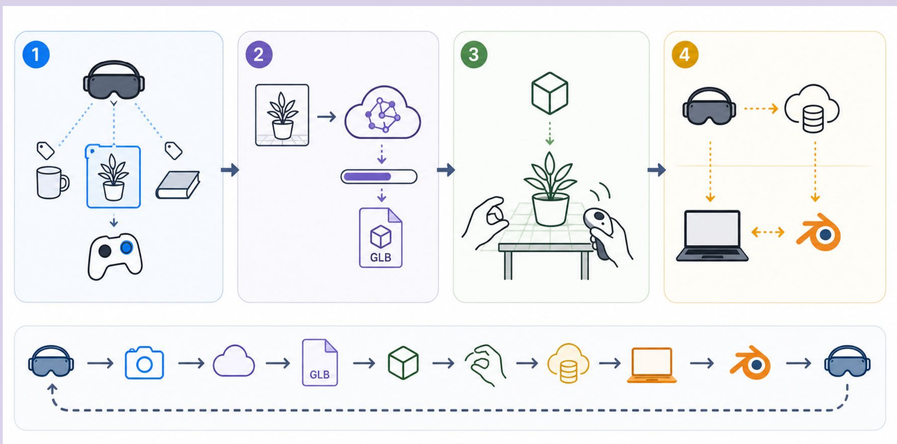


Fig 1.3: Workflow of the implementation

## Results

4.1 Functional Outcomes

- **End-to-end capture-to-grab in under 30 seconds** on Meta Quest 3 hardware.
- **Zero manual modelling steps** required from the user.
- **Bidirectional round-trip** completes in <10 seconds (excluding human edit time in Blender).
- **Multiple simultaneous models** coexist in the scene without state conflicts.
- **Hand tracking, controller grabbing, *and* distance grabbing** all function on the same spawned objects.

## Conclusion

- This work demonstrates that the long-promised "seamless" continuum between real and virtual is achievable on consumer hardware *if* the design treats the human, not the machine, as the centre of the loop. By committing to gaze-driven selection, hand-only manipulation, and round-trip authoring, we collapsed a workflow that traditionally requires multiple applications and expert skill into a single continuous gesture sequence.
- The implications extend beyond a research artefact. The same loop applies wherever a user needs to bring physical objects into a digital workflow without losing presence: museums, classrooms, design studios, surgical training, factory floors. The technical components — passthrough cameras, neural detectors, generative 3D models, hand tracking — are now commodity. The remaining problem is one of **interaction design**, and the result here suggests that getting the loop right is more important than getting any single component perfect.

## References

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